Notes

CAFFEINE & CADAVER
MBBS BIOCHEMISTRY NOTES SERIES

Protein Chemistry & Haemoglobin

MBBS BIOCHEMISTRY · PAPER 1

1° – 4° Structure | α-Helix · β-Sheet | Hb vs Mb | O_2 Dissociation | 2,3-BPG · Bohr Effect | HbS · HbF · HbA | Haem Degradation | Bilirubin Metabolism | Plasma Proteins | Electrophoresis

01 Levels of Protein Structure

4 levels exist. **Primary → Secondary → Tertiary → Quaternary**. Each higher level depends on the one below it.

1°
Primary Structure
Peptide Bond (Covalent)
Linear sequence of amino acids. Determined by the gene. **Sanger's method** identifies it.
Example: Insulin (51 AA)

2°
Secondary Structure
H-bonds (backbone NH···CO)
Regular repeating patterns: **α-helix**, **β-pleated sheet**, β-turns. Stabilised by hydrogen bonds between backbone atoms.

3°
Tertiary Structure
Hydrophobic + H-bond + IonicBond + Disulphide
3D folding of a single polypeptide. Hydrophobic core. Disulphide bonds (covalent). Determines biological function.

4°
Quaternary Structure
Non-covalent + some S-S
Assembly of 2+ polypeptide subunits. Example: **Haemoglobin ($\alpha_2\beta_2$)**. Not all proteins have this level.

MNEMONIC — BONDS BY LEVEL

"Pretty Smart Tertiary Queens" → Peptide, Stable H-bonds, Three types (hydrophobic+H+ionic+S-S), Quaternary = all of the above

Or simpler: **1° Peptide → 2° Hydrogen → 3° Multiple (HISD: Hydrophobic, Ionic, S-S, Disulphide) → 4° Same as 3° but between chains**

CLINICAL CORRELATION — WHY PRIMARY STRUCTURE MATTERS

A single amino acid substitution in primary structure → disease. In **Sickle Cell Anaemia**, Glu→Val at position 6 of β-globin chain creates HbS. This demonstrates that *primary structure determines function* — a core biochemistry principle.

FORCES STABILISING PROTEIN STRUCTURE

FORCE	TYPE	WHERE IT ACTS	DISRUPTED BY
Peptide bond	Covalent	1° (backbone)	Proteases
H-bond	Non-covalent	2°, 3°	Urea, heat
Hydrophobic interactions	Non-covalent	3°	Detergents, organic solvents
Ionic/electrostatic	Non-covalent	3°	pH change, salt
Disulphide bond (-S-S-)	Covalent	3° (surface/interior)	Reducing agents (β-ME, DTT)
Van der Waals	Non-covalent	3°, 4°	Weak, easily disrupted

⚠️ **Denaturation** disrupts 2°, 3°, 4° structure but NOT 1° (peptide bonds remain intact). Renaturation is possible if 1° is intact.

02 α-Helix & β-Pleated Sheet

A-HELIX ⭐ EXAMINER FAV

- ✦ Right-handed helix (most proteins)
- ✦ **3.6 amino acids** per turn
- ✦ Rise per residue: **1.5 Å**
- ✦ Pitch (rise per turn): **5.4 Å**

B-PLEATED SHEET PYQ 2017, 2018

- ✦ Extended, zigzag conformation
- ✦ Two types: **Parallel** and **Antiparallel**
- ✦ H-bonds run **perpendicular to chain direction**

✦ H-bonds: between **NH of residue n** and **C=O of residue n-4**

✦ H-bonds run **parallel to helix axis**

✦ R-groups point **outward** from helix

✦ Disrupted by: Proline (no NH), Glycine (too flexible), charged R-groups

EXAMPLE PROTEINS

Keratin (hair, nails): nearly entirely α-helical, coiled-coil structure

Myoglobin: 8 α-helical segments (A–H)

✦ Antiparallel: stronger H-bonds (more stable)

✦ R-groups alternate: above and below sheet plane

✦ Rise per residue: **3.5 Å**

FEATURE	PARALLEL	ANTIPARALLEL
Chain direction	Same	Opposite
H-bond angle	Bent	Linear
Stability	Less	More

FIBROUS VS GLOBULAR PROTEINS

PROPERTY	FIBROUS (STRUCTURAL)	GLOBULAR (FUNCTIONAL)
Shape	Long, rod-like	Spherical/compact
Solubility	Insoluble in water	Soluble
Dominant 2° structure	α-helix (keratin) or β-sheet (fibroin)	Mix of helix, sheet, coil
Role	Structure (hair, tendons, silk)	Enzyme, transport, Hb, Ab
Example	Keratin, Collagen, Fibroin	Haemoglobin, Myoglobin, Lysozyme

03 Haemoglobin vs Myoglobin

FEATURE	HAEMOGLOBIN (HB)	MYOGLOBIN (MB)
Location	RBCs (blood)	Muscle cells
Subunits	Tetramer: **$\alpha_2\beta_2$** (HbA)	Monomer (1 chain)
MW	~68,000 Da	~17,000 Da
Haem groups	4 (one per subunit)	1
O_2 affinity curve	**Sigmoidal (S-shaped)**	**Hyperbolic**
Cooperativity	Yes (allosteric)	No (no cooperativity)
Bohr effect	**Yes** (CO_2 & H^+ ↓ affinity)	None
2,3-BPG effect	**Yes** (↓ O_2 affinity)	None
Function	O_2 **transport**	O_2 **storage**
P_{50} (pO_2 at 50% saturation)	~26 mmHg (higher = lower affinity)	~2 mmHg (lower = higher affinity)
T-state / R-state	T (tense/deoxy), R (relaxed/oxy)	N/A (monomer)
Secondary structure	8 α-helical segments per chain	8 α-helical segments (A–H)

HAEM GROUP STRUCTURE

Haem is a **porphyrin ring** (protoporphyrin IX) containing a central **Fe^{2+}** ion.

▶ HAEM ARCHITECTURE

Core	→	Central Atom	→	Bonds
Protoporphyrin IX 4 pyrrole rings bridged by methine		Fe^{2+} 6 coordination bonds		4 to N of pyrroles 1 to proximal His (F8) 1 to O_2 or distal His (E7)

Fe^{2+} (ferrous) binds O_2. Oxidation to Fe^{3+} (ferric) = methaemoglobin — cannot carry O_2.

T-STATE VS R-STATE (ALLOSTERIC BEHAVIOUR) ⭐ HIGH YIELD

▶ HB CONFORMATIONAL SWITCH

T-State (Tense)
Deoxyhaemoglobin
Low O_2 affinity · Stabilised by 2,3-BPG, H^+, CO_2, salt bridges

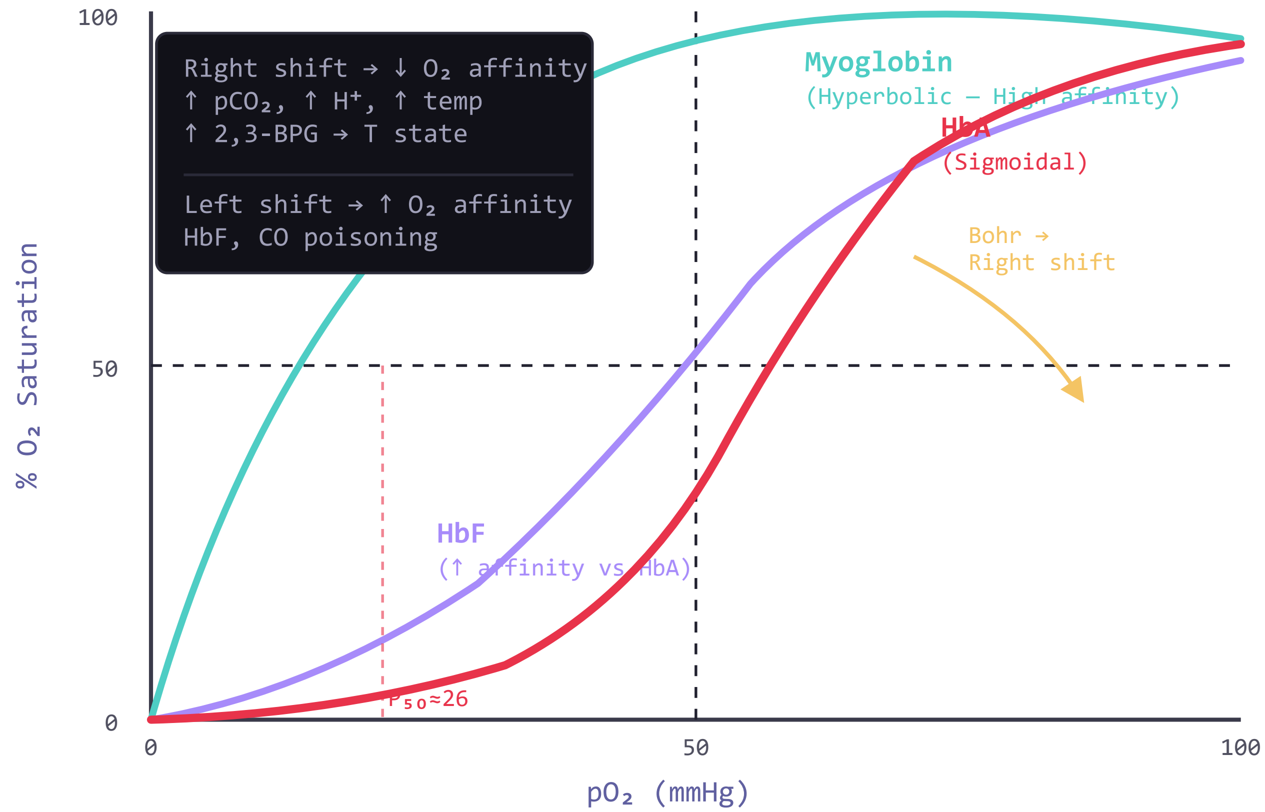
⇌

R-State (Relaxed)
Oxyhaemoglobin
High O_2 affinity · Favoured on O_2 binding

As first O_2 binds → conformational change → subsequent binding becomes EASIER → **cooperative binding** → Sigmoidal curve.

04 O_2 Dissociation Curve

▶ OXYGEN DISSOCIATION CURVES — HB VS MB VS HBF



RIGHT SHIFT → ↓ O_2 AFFINITY (O_2 UNLOADING)

LEFT SHIFT → ↑ O_2 AFFINITY (O_2 HOARDING)

✦ ↑ pCO_2 (exercise, tissues)
✦ ↑ H^+ (acidosis) — Bohr effect
✦ ↑ Temperature
✦ ↑ 2,3-BPG

Right shift = **MORE O_2 delivered to tissues**. Good in exercise, fever.

✦ ↓ pCO_2 (hyperventilation)
✦ ↓ H^+ (alkalosis)
✦ ↓ Temperature
✦ ↓ 2,3-BPG (stored blood)
✦ HbF (foetal Hb)
✦ CO poisoning

Left shift = **Hb grabs O_2 but won't release it**. Dangerous in CO poisoning.

05 2,3-BPG & Bohr Effect

2,3-BISPHOSPHOGLYCERATE (2,3-BPG) ⭐ PYQ EVERY YEAR

▶ RAPOPORT-LUEBERING CYCLE (HOW 2,3-BPG IS MADE)

1,3-BPG (Glycolysis intermediate) → Bisphosphoglycerate mutase **2,3-BPG** Only in RBCs (no mitochondria) → 3-PG Continues glycolysis

Mechanism of Action:

✦ 2,3-BPG binds the **central cavity** of Hb between β-chains
✦ Only binds **T-state (deoxyHb)**
✦ Stabilises T-state → ↓ O_2 affinity → right shift
✦ **1 molecule** of 2,3-BPG per 1 Hb molecule

Clinical Significance:

✦ **High altitude:** ↑ 2,3-BPG → ↑ O_2 delivery to tissues
✦ **Stored blood:** 2,3-BPG depletes → ↑ O_2 affinity → poor O_2 delivery (transfusion problem)
✦ **Anaemia:** ↑ 2,3-BPG (compensatory)
✦ **HbF:** doesn't bind 2,3-BPG (γ-chains) → ↑ affinity → foetal O_2 extraction from maternal Hb

BOHR EFFECT PYQ 2013, MULTIPLE ⭐

Definition: The decrease in O_2 affinity of Hb caused by **↑ CO_2 and/or ↑ H^+**. Facilitates O_2 delivery to metabolically active tissues.

▶ BOHR EFFECT MECHANISM

↑ CO_2 in tissues → $CO_2 + H_2O$ → H_2CO_3 → **H^+ + HCO_3^-** (by carbonic anhydrase)

↓

H^+ binds to **Histidine residues** on Hb (β-146 His, α-122 His) → forms salt bridges → stabilises **T-state**

↓

↓ O_2 affinity → O_2 released to tissues

↓

In lungs: ↓ CO_2 → salt bridges break → R-state → O_2 loads onto Hb again

KEY CONCEPT

Bohr = "H goes in → O_2 goes out" (tissues); "O_2 goes in → H goes out" (lungs)

06 HbA · HbS · HbF

HB TYPE	STRUCTURE	ELECTROPHORESIS	O_2 AFFINITY	CLINICAL
HbA (Adult)	$\alpha_2\beta_2$	Moves fastest toward anode at pH 8.6	Normal (P_{50} 26 mmHg)	Normal adult haemoglobin
HbA_2	$\alpha_2\delta_2$	Slow band	Similar to HbA	↑ in β-Thalassaemia (2–3%)
HbF (Foetal)	$\alpha_2\gamma_2$	Faster than HbS but slower than HbA	**↑ (P_{50} ~20 mmHg)**	↑ O_2 affinity; normal until 6 months; ↑ in β-Thal, sickle cell
HbS (Sickle)	$\alpha_2\beta_2^S$	Moves toward cathode (different from HbA)	Normal when oxygenated	Glu→Val at β6; polymerises when deoxygenated → sickling
HbM	Various α or β mutations at proximal/distal His	Abnormal	Low (Fe^{3+} locked)	Methaemoglobinaemia
HbA_1c (Glycated)	HbA + glucose at β-N-terminal Val	Moves ahead of HbA	Slightly ↓	**Diabetes monitoring** (reflects 8–12 wk glucose)

SICKLE CELL ANAEMIA — MOLECULAR BASIS ⭐ PYQ REPEATEDLY

▶ HBS PATHOPHYSIOLOGY

Point mutation: **GAG** → **GTG** (codon 6, β-globin gene)

↓

Amino acid change
Glutamate (charged, hydrophilic) → **Valine (non-polar, hydrophobic)**

↓

Creates a **"sticky patch"** on deoxyHbS surface
Val fits into complementary hydrophobic pocket on adjacent Hb molecule

↓

Polymerisation of deoxyHbS → long fibres → **Sickling** of RBCs

↓

Haemolytic anaemia + Vaso-occlusion + Organ damage (spleen, kidney, brain)

WHY HBF IS PROTECTIVE IN SICKLE CELL

HbF ($\alpha_2\gamma_2$) does NOT polymerise with HbS. High HbF levels (as in neonates, or with hydroxyurea treatment) **inhibit sickling** by diluting HbS polymer formation. This is why symptoms appear only after HbF switches to HbA (~6 months of age).

⚠️ **EQ alert:** "Patient with HbS often suffers from anaemia" — answer: haemolytic due to sickling + membrane damage. Sickled cells are rigid and get destroyed in spleen.

EQ TEMPLATE — HBF HAS MORE O_2 AFFINITY THAN HBA

> *"HbF has more affinity towards oxygen than HbA." [2019-S]*

1. **Fact:** HbF has P_{50} ~20 mmHg vs HbA P_{50} ~26 mmHg → lower P_{50} = higher O_2 affinity
2. **Structural Basis:** HbF has γ-chains instead of β-chains. γ-chains have Serine at position 143 instead of Histidine (β-143 His)
3. **2,3-BPG Interaction:** 2,3-BPG binds poorly to γ-chains (Ser 143 creates steric hindrance). Less 2,3-BPG effect → HbF stays in R-state more → ↑ O_2 affinity
4. **Physiological Significance:** Foetus extracts O_2 from maternal HbA across the placenta. HbF's higher affinity ensures O_2 transfer from maternal to foetal blood
5. **Clinical:** After birth, γ→β switch occurs; HbF → HbA by 6 months. Persistence of HbF (HPFH) is actually protective in sickle cell and β-thalassaemia

07 Haem Degradation

FIG 4.1: HAEM DEGRADATION PATHWAY

Aged RBCs (80%) + Cytochromes (20%)
Destroyed in RES (Spleen, Macrophages)

↓ Globin removed

Haem

↓ Haem oxygenase (Needs O_2, NADPH)

Biliverdin (Green)
Porphyrin ring opened

↓ Biliverdin reductase (NADPH)

Bilirubin (Yellow) — Unconjugated
Lipid-soluble · Toxic · Binds Albumin

↓ UDP-Glucuronosyltransferase (Liver)
+ 2 UDP-Glucuronic Acid

Bilirubin Diglucuronide — Conjugated
Water-soluble · Non-toxic (In Liver)

↓ Excreted in bile → Intestine
(Bacterial action)

Urobilinogen → Stercobilin (Brown)

🧠 MNEMONIC — HAEM BREAKDOWN COLOURS

"Haem Goes Bye-bye, Urine Stays" = Haem → Green (biliverdin) → Bilirubin (yellow) → Urobilinogen → Stercobilin

08 Bilirubin & *Jaundice*

PARAMETER	HAEMOLYTIC (PRE-HEPATIC)	HEPATOCELLULAR	OBSTRUCTIVE (POST-HEPATIC)
Serum Unconjugated Bili	↑↑↑	↑	Normal/slightly ↑
Serum Conjugated Bili	Normal	↑	↑↑↑
Urine Bilirubin	Absent (indirect insoluble)	Present	**Present** ↑↑
Urine Urobilinogen	↑↑↑	↑	↓/Absent
Stool colour	Dark (↑ stercobilin)	Pale	**Clay/pale (no bile)**
Urine colour	Dark (urobilinogen)	Dark	Dark (bilirubin)
ALT/AST	Normal	↑↑↑	Normal/↑
Alkaline Phosphatase	Normal	↑	↑↑↑
Example	G6PD deficiency, Thalassaemia, RBC destruction	Hepatitis, Cirrhosis	Gallstones, Cholestasis, CA head of pancreas

NEONATAL/PHYSIOLOGICAL JAUNDICE PYQ 2011, 2017, 2021

Why it Occurs:

✦ HbF → HbA switch: massive RBC turnover

✦ Immature UGT1A1 in neonatal liver → poor conjugation

✦ Increased enterohepatic circulation (intestinal bacteria absent)

✦ Albumin less, so less binding → free bili crosses BBB → **Kernicterus**

Phototherapy Mechanism:

✦ Blue light (460 nm) → converts bilirubin (4Z,15Z) → photoisomers (4Z,15E lumirubin)

✦ Photoisomers are **water-soluble** → can be excreted in bile/urine WITHOUT conjugation

✦ Works without needing UGT1A1 → ideal for neonates

✦ Preferred over phenobarbitone in Crigler-Najjar Type I (no UGT at all)

VAN DEN BERGH REACTION PYQ 2019

Direct Reaction
= Conjugated bilirubin reacts DIRECTLY with diazo reagent (no alcohol needed)
= Water-soluble

Indirect Reaction
= Unconjugated bilirubin reacts ONLY after adding alcohol (breaks albumin bond)
= Lipid-soluble

Total bilirubin = Direct + Indirect. Normal: Total <1 mg/dL, Direct <0.3 mg/dL.

09 Plasma *Proteins*

Normal Total Plasma Protein: **6–8 g/dL** . Albumin: **3.5–5 g/dL** . A:G ratio: **1.5–2.5:1**

PROTEIN	SITE OF SYNTHESIS	FUNCTION	CLINICAL SIGNIFICANCE
Albumin	Liver	Oncotic pressure, transport (bilirubin, FA, drugs, Ca^{2+})	↓ in liver disease, nephrotic syndrome → oedema
α_1-Antitrypsin	Liver	Inhibits elastase (neutrophil)	Deficiency → emphysema, liver disease
α_2-Macroglobulin	Liver	Broad-spectrum protease inhibitor	↑ in nephrotic syndrome (too large to be lost)
Transferrin	Liver	Iron transport (Fe^{3+} binding)	↓ in liver disease, malnutrition; ↑ in iron deficiency
Haptoglobin	Liver	Binds free Hb → prevents renal damage	↓ **in haemolysis** (consumed)
Ceruloplasmin	Liver	Copper transport (6–7 Cu atoms), ferroxidase activity	↓ **in Wilson's disease**; Acute phase reactant
Fibrinogen	Liver	Coagulation (→ fibrin by thrombin)	↑ in inflammation; Acute phase reactant
Immunoglobulins	Plasma cells (B cells)	Antibody function (IgG, IgA, IgM, IgD, IgE)	↑ in infections; ↓ in immunodeficiency; M-spike in myeloma
Prothrombin	Liver (Vit K-dependent)	Clotting factor II (→ thrombin)	↓ in Vit K deficiency, liver disease
C-Reactive Protein (CRP)	Liver	Acute phase reactant; opsonin	↑ in infection, inflammation, MI

ACUTE PHASE REACTANTS

PYQ 2014, 2022

Positive APR (↑ in inflammation):

- ✦ CRP, Serum Amyloid A
- ✦ Fibrinogen, Prothrombin
- ✦ Ceruloplasmin, Haptoglobin
- ✦ α_1-Antitrypsin, α_2-Macroglobulin
- ✦ Ferritin, Complement proteins

Negative APR (↓ in inflammation):

- ✦ **Albumin**
- ✦ Transferrin
- ✦ Transthyretin (Prealbumin)
- ✦ Retinol-binding protein

Negative APRs are liver "luxury proteins" — synthesis switches to positive APRs during stress.

📝 EQ TEMPLATE — HYPOALBUMINAEMIA LEADS TO PERIPHERAL OEDEMA [2025]

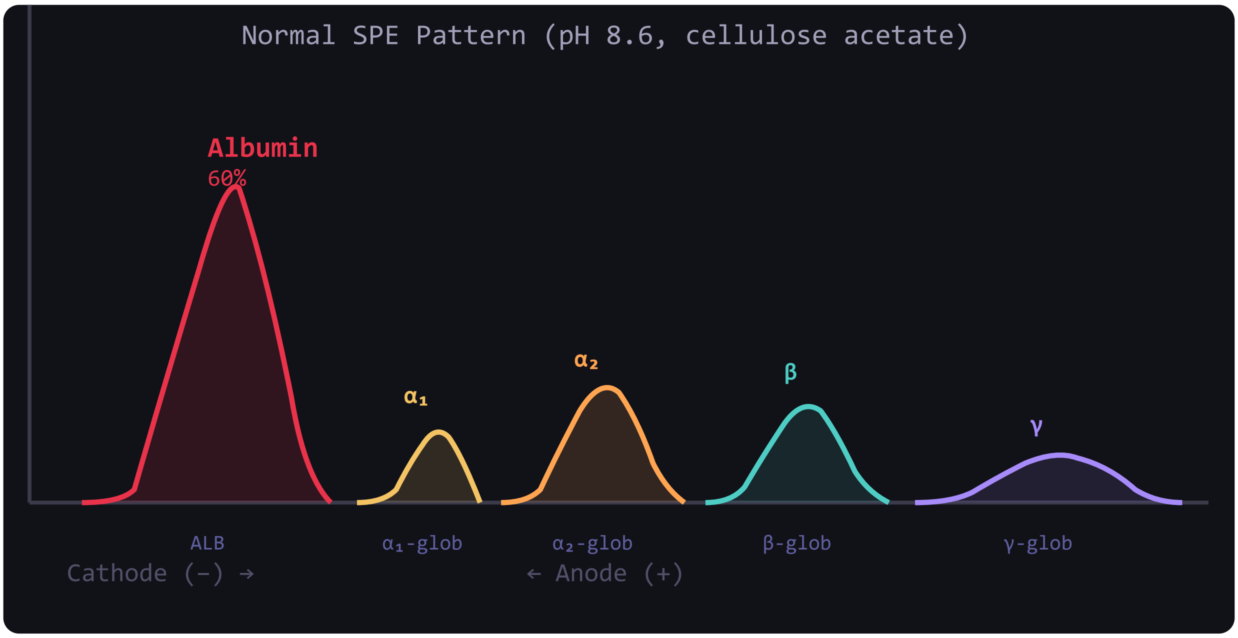
> *"Hypoalbuminaemia leads to peripheral oedema."*

1. **Normal role of albumin:** Albumin (3.5–5 g/dL) is the principal contributor to **plasma oncotic (colloid osmotic) pressure** (~25 mmHg)
2. **Oncotic pressure function:** Draws water BACK into capillaries from interstitial space (opposing hydrostatic pressure that pushes fluid out)
3. **Hypoalbuminaemia:** ↓ plasma oncotic pressure → hydrostatic pressure > oncotic pressure → net fluid filtration into interstitium
4. **Result:** Peripheral oedema (pitting), ascites, pulmonary oedema, periorbital oedema
5. **Causes:** Liver disease (↓ synthesis), Nephrotic syndrome (↑ loss in urine), Malnutrition (kwashiorkor), Malabsorption
6. **Example:** Nephrotic syndrome — serum albumin <2.5 g/dL → massive generalised oedema (anasarca)

10 Serum Protein *Electrophoresis*

⚡ At pH 8.6 (alkaline), all serum proteins are **negatively charged** → migrate toward anode (+ve). Separation based on size and charge.

▶ SERUM PROTEIN ELECTROPHORESIS PATTERN — NORMAL VS PATHOLOGICAL



BAND	PROTEINS INCLUDED	NORMAL %	↑ IN	↓ IN
Albumin	Albumin only	55–65%	Dehydration	Liver disease, nephrotic, malnutrition, inflammation
α_1-globulin	α_1-Antitrypsin, α_1-glycoprotein	2–4%	Acute inflammation, pregnancy	α_1-Antitrypsin deficiency
α_2-globulin	Haptoglobin, Ceruloplasmin, α_2-Macroglobulin	6–12%	Nephrotic syndrome (↑ α_2-Macro)	Haemolysis (↓ haptoglobin)
β-globulin	Transferrin, β-Lipoprotein, Complement (C3, C4)	8–14%	Iron deficiency (↑ transferrin), Obstructive jaundice	Liver failure
γ-globulin	IgG (predominant), IgA, IgM, IgD, IgE	12–22%	Infections, autoimmune diseases; **M-spike** in myeloma	Immunodeficiency, nephrotic syndrome

Pathological SPE Patterns

Multiple Myeloma: Narrow, tall "M-spike" (monoclonal Ig) in γ or β region. ↓ normal Igs (immunoparesis).

Cirrhosis: ↓ Albumin + β-γ bridge (IgA bridges β and γ zones). ↑ γ polyclonal.

Nephrotic Syndrome: ↓↓ Albumin + ↑ α_2-globulin (α_2-Macroglobulin too big to be lost through GBM) + ↓ γ.

Acute Inflammation: ↓ Albumin + ↑ α_1 + ↑ α_2 (acute phase reactants). Fibrinogen absent in serum but present in plasma.

MODULE 04 · PROTEIN CHEMISTRY & HAEMOGLOBIN

Caffeine & Cadaver · MBBS Biochemistry Notes · 2010-2025

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